

Output Code of Project "Performance Benchmarking of CNN and Random Forest Classifiers on the Fashion-MNIST Dataset"

Code from Windows Powershell

June 24, 2026

Course: Computer Vision DLBAIPCV01_CF

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Course of Study: Applied Artificial Intelligence (B.Sc.)

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Windows PowerShell

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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

```
PS C:\Users\theae> cd C:\Users\theae\Desktop\fashion_mnist_project
```

```
PS C:\Users\theae\Desktop\fashion_mnist_project> fashion_env\Scripts\activate
```

```
(fashion_env) PS C:\Users\theae\Desktop\fashion_mnist_project> python fashion_classifier.py
```

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

I0000 00:00:1781295603.821218 37168 port.cc:153] oneDNN custom operations are on. You may see slightly

↪ different numerical results due to floating-point round-off errors from different computation orders. To turn
↪ them off, set the environment variable `TF\_ENABLE\_ONEDNN\_OPTS=0`.

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

I0000 00:00:1781295606.792046 37168 port.cc:153] oneDNN custom operations are on. You may see slightly

↪ different numerical results due to floating-point round-off errors from different computation orders. To turn
↪ them off, set the environment variable `TF\_ENABLE\_ONEDNN\_OPTS=0`.

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FASHION-MNIST CLASSIFICATION: CNN vs RANDOM FOREST

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[1/7] Loading Fashion-MNIST dataset...

Total training images: 60000

Test images: 10000

Image size: 28x28 pixels

Number of classes: 10

Classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot

[2/7] Preprocessing data...

Pixel normalization: [0,255] -> [0,1]

Train set: 48000 images

Validation set: 12000 images

Test set: 10000 images

CNN input shape: (48000, 28, 28, 1)

RF input shape: (48000, 784)

[3/7] Building and training CNN...

I0000 00:00:1781295608.110355 37168 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use

↪ available CPU instructions in performance-critical operations.

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow

↪ with the appropriate compiler flags.

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if

↪ CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

CNN Architecture:

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496

max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
conv2d_2 (Conv2D)	(None, 3, 3, 64)	36,928
flatten (Flatten)	(None, 576)	0
dense (Dense)	(None, 64)	36,928
dense_1 (Dense)	(None, 10)	650

Total params: 93,322 (364.54 KB)
 Trainable params: 93,322 (364.54 KB)
 Non-trainable params: 0 (0.00 B)

Training CNN (10 epochs)...

Epoch 1/10
 1500/1500 11s 7ms/step - accuracy: 0.8085 - loss: 0.5245 - val_accuracy: 0.8559 - val_loss: 0.3895
 Epoch 2/10
 1500/1500 10s 7ms/step - accuracy: 0.8768 - loss: 0.3373 - val_accuracy: 0.8851 - val_loss: 0.3309
 Epoch 3/10
 1500/1500 10s 6ms/step - accuracy: 0.8953 - loss: 0.2875 - val_accuracy: 0.8971 - val_loss: 0.2878
 Epoch 4/10
 1500/1500 10s 6ms/step - accuracy: 0.9049 - loss: 0.2571 - val_accuracy: 0.8953 - val_loss: 0.2888
 Epoch 5/10
 1500/1500 10s 7ms/step - accuracy: 0.9129 - loss: 0.2303 - val_accuracy: 0.9066 - val_loss: 0.2656
 Epoch 6/10
 1500/1500 10s 6ms/step - accuracy: 0.9241 - loss: 0.2069 - val_accuracy: 0.9079 - val_loss: 0.2635
 Epoch 7/10
 1500/1500 10s 6ms/step - accuracy: 0.9303 - loss: 0.1885 - val_accuracy: 0.9082 - val_loss: 0.2532
 Epoch 8/10
 1500/1500 9s 6ms/step - accuracy: 0.9368 - loss: 0.1706 - val_accuracy: 0.9090 - val_loss: 0.2584
 Epoch 9/10
 1500/1500 10s 7ms/step - accuracy: 0.9420 - loss: 0.1561 - val_accuracy: 0.9119 - val_loss: 0.2622
 Epoch 10/10
 1500/1500 9s 6ms/step - accuracy: 0.9471 - loss: 0.1399 - val_accuracy: 0.9125 - val_loss: 0.2680

CNN training completed in 98.56 seconds

[4/7] Building and training Random Forest...

Configuration: n_estimators=100, criterion='entropy', max_depth=100

Random Forest training completed in 11.08 seconds

[5/7] Getting predictions...

Predictions completed

[6/7] Generating results...

TABLE 1: PRECISION AND RECALL PER CLASS (TEST SET)

Class	CNN Prec	CNN Rec	RF Prec	RF Rec
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T-shirt/top	0.852	0.876	0.821	0.864
Trouser	0.994	0.980	0.994	0.964
Pullover	0.850	0.883	0.762	0.787
Dress	0.923	0.902	0.868	0.907
Coat	0.812	0.910	0.758	0.806
Sandal	0.987	0.973	0.980	0.961
Shirt	0.784	0.675	0.718	0.583
Sneaker	0.948	0.975	0.934	0.955
Bag	0.984	0.965	0.955	0.973
Ankle boot	0.971	0.964	0.949	0.949

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TABLE 2: PRECISION AND RECALL PER CLASS (TRAINING SET)

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Class	CNN Prec	CNN Rec	RF Prec	RF Rec

T-shirt/top	0.916	0.937	1.000	1.000
Trouser	0.999	0.992	1.000	1.000
Pullover	0.926	0.943	1.000	1.000
Dress	0.972	0.959	1.000	1.000
Coat	0.894	0.964	1.000	1.000
Sandal	0.999	0.994	1.000	1.000
Shirt	0.912	0.828	1.000	1.000
Sneaker	0.979	0.989	1.000	1.000
Bag	0.999	0.992	1.000	1.000
Ankle boot	0.988	0.983	1.000	1.000

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TABLE 3: TRAINING TIMES

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CNN: 98.56 seconds (1.64 minutes)

Random Forest: 11.08 seconds (0.18 minutes)

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TABLE 4: ACCURACY SUMMARY

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Model	Train	Validation	Test

CNN	0.9581 (95.81%)	0.9125 (91.25%)	0.9103 (91.03%)
Random Forest	1.0000 (100.00%)	0.8852 (88.52%)	0.8749 (87.49%)

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GENERATING CONFUSION MATRIX IMAGES

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Saved: RF_train_confusion_matrix.png

Saved: RF_test_confusion_matrix.png

Saved: CNN_train_confusion_matrix.png

Saved: CNN_test_confusion_matrix.png

Saving CSV files...

Saved: RF_train_confusion_matrix.csv

Saved: RF_test_confusion_matrix.csv
Saved: CNN_train_confusion_matrix.csv
Saved: CNN_test_confusion_matrix.csv

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WORST PERFORMING CATEGORIES ANALYSIS

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CNN - 3 worst performing categories (by precision on test set):

6: Shirt - Precision: 0.784, Recall: 0.675
4: Coat - Precision: 0.812, Recall: 0.910
2: Pullover - Precision: 0.850, Recall: 0.883

Random Forest - 3 worst performing categories (by precision on test set):

6: Shirt - Precision: 0.718, Recall: 0.583
4: Coat - Precision: 0.758, Recall: 0.806
2: Pullover - Precision: 0.762, Recall: 0.787

EXPLANATION FOR LOW PERFORMANCE:

- Shirts, T-shirts/tops, Pullovers, and Coats are frequently confused
- These are all upper-body garments with sleeves and similar silhouettes
- Grayscale 28x28 images lack texture and color details to distinguish them
- A shirt vs a T-shirt vs a Pullover can look nearly identical at low resolution
- The RF classifier struggles more because flattening removes spatial relationships

BEST PERFORMING CATEGORIES:

- Trousers, Bags, Sandals, Sneakers, Ankle boots have distinctive shapes
- These are less likely to be confused with other categories

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COMPARISON AND RECOMMENDATION

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MODEL COMPARISON:

Test Accuracy: CNN = 91.03% vs RF = 87.49%
Training time: CNN = 98.6s vs RF = 11.1s
Advantage CNN: +3.5% accuracy improvement
Advantage RF: 0.1x faster training

RECOMMENDATION: USE CNN IN PRODUCTION

JUSTIFICATION:

1. Higher Accuracy: CNN achieves higher test accuracy than RF
2. Better Generalization: CNN maintains smaller gap between train and test
3. Feature Learning: CNNs learn spatial hierarchies (edges, textures, shapes)
4. Pixel Independence: RF treats each pixel separately, missing spatial relationships
5. Fashion items have consistent shapes - CNNs exploit this structure naturally
6. The accuracy gain justifies the longer training time for production

WHEN TO USE RANDOM FOREST INSTEAD:

- Extremely limited computational resources
- Need very fast training (< 1 minute)
- No GPU available and CPU is very slow
- Simple baseline for comparison required

```
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SCRIPT COMPLETED SUCCESSFULLY
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```

OUTPUT FILES GENERATED:

```
CNN_test_confusion_matrix.png
CNN_train_confusion_matrix.png
RF_test_confusion_matrix.png
RF_train_confusion_matrix.png
CNN_test_classification_report.csv
CNN_train_classification_report.csv
RF_test_classification_report.csv
RF_train_classification_report.csv
cnn_test_confusion_matrix.csv
cnn_train_confusion_matrix.csv
rf_test_confusion_matrix.csv
rf_train_confusion_matrix.csv
```

```
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All requirements from Task 2 have been satisfied:
  Fashion-MNIST dataset description
  CNN implementation (TensorFlow/Keras)
  Random Forest implementation
  4 confusion matrices (train + test)
  Precision/Recall tables (train + test)
  Training times reported
  Worst classes analysis with explanation
  Comparison and recommendation
=====
```

```
(fashion_env) PS C:\Users\theae\Desktop\fashion_mnist_project>
```